
AfriVTON-Bench: Evaluating the Robustness of Virtual Try-On Models to African Textile Patterns and Garment Structures

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Abstract

Virtual Try-On (VTON) systems have advanced considerably, yet their generalisation across global fashion diversity remains constrained by training corpora that skew heavily toward Western garment aesthetics. We introduce AfriVTON-Bench, a benchmark designed to evaluate VTON robustness on African garments—a domain characterised by high-frequency repeating wax-print and woven patterns as well as voluminous non-Western draping silhouettes that lie far outside the distribution of existing evaluation suites. We curate 111 African garment images spanning 16 categories across 7 countries and 5 sub-regions, pair them with a 50-image Western control set drawn from VITON-HD, and evaluate two state-of-the-art architectures across 1,012 person–garment pairs. Both models exhibit pronounced garment–person reconstruction asymmetry and a structural fidelity deficit of 9–10 percentage points in gSSIM on African garments relative to Western controls. We identify three systematic failure modes—pattern dissolution, drape collapse, and tonal drift—that cut across both architectures, indicating that the root cause lies in the shared Western training distribution rather than any particular design choice. Beyond documenting failures, AfriVTON-Bench demonstrates how culturally grounded benchmarks can surface generalisation gaps invisible to standard evaluation, articulating a constructive path toward more culturally inclusive generative AI.

1. Introduction

Virtual Try-On (VTON) systems, which digitally composite a target garment onto a reference person image, have undergone rapid capability gains driven by diffusion-based architectures. Systems such as OOTDiffusion (Xu et al., 2024), IDM-VTON (Choi et al., 2024), CatVTON (Chong et al., 2025), and FitDiT (Jiang et al., 2024) now produce photorealistic synthesis results on the VITON-HD benchmark (Choi et al., 2021), with perceptual metrics approaching those of ground-truth photograph pairs. These capabilities have been deployed in online fashion retail, where try-on tools reduce return rates by enabling customers to visualise fit without physical garments.

The apparent quality gains, however, are tightly coupled to a narrow training distribution. VITON-HD (Choi et al., 2021) and DressCode (Morelli et al., 2023b), the two benchmarks on which virtually all contemporary models are trained and evaluated, were curated from Western fashion e-commerce platforms. The garments in these datasets are overwhelmingly Western in silhouette and lightly textured in fabric, and the associated subjects cluster in the lighter half of perceptual skin tone scales. Published performance figures may therefore substantially overestimate system quality for users and garment types outside this distribution. This is a well-known instance of the dataset bias problem formalised by Buolamwini & Gebru (2018), who showed that aggregate benchmark performance routinely conceals systematic degradation across underrepresented subgroups.

African textile garments pose a structurally distinct challenge for existing VTON pipelines. Fabrics such as Ankara, Kente, Adire, and Aso-Oke carry high-frequency repeating motifs with strong geometric regularity and high local contrast—properties that expose known failure modes in both spatial warping and diffusion synthesis stages (Jiang et al., 2024). Traditional garments including Agbada, Boubou, and Dashiki have voluminous, non-fitted silhouettes that differ substantially from the body-congruent clothing dominant in existing benchmarks. Work on African fashion in computer vision remains sparse. AFRIFASHION1600 (Oyewusi et al., 2021) targets classification rather than generation, AFRIGAN (Salami et al., 2021) addresses unconditional texture synthesis, and Omotayo et al. (2024)

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report that fewer than 0.06% of computer vision papers address African-specific visual domains. No existing VTON work examines model behaviour on African textiles, garment structures, or the skin tone distribution of African subjects.

This paper contributes to the *Culture* $\times$ *AI* discourse by demonstrating that culturally grounded benchmarks are not merely fairness exercises but essential tools for surfacing generalisation failures of broader relevance to the field. Specifically, we contribute: (1) AfriVTON-Bench, a curated benchmark spanning 16 African garment categories paired with subjects covering the full Monk Skin Tone scale; (2) a stratified robustness evaluation of two architecturally distinct VTON models; and (3) a characterisation of three systematic, architecture-agnostic failure modes that implicate the Western centrality of training data as the root cause.

2. Related Work

Virtual Try-On Architectures and Benchmarks. Early neural VTON approaches used thin-plate spline or appearance flow transformations (Han et al., 2018; Wang et al., 2018). Diffusion-based architectures subsequently raised synthesis quality: LaDI-VTON (Morelli et al., 2023a), OOT-Diffusion (Xu et al., 2024), IDM-VTON (Choi et al., 2024), CatVTON (Chong et al., 2025), and FitDiT (Jiang et al., 2024) each introduce distinct conditioning strategies. The benchmark landscape remains narrow: VITON-HD and DressCode share Western provenance and have not been audited for pattern complexity or skin tone composition.

Bias in Generative Vision. Buolamwini & Gebru (2018) formalised stratified auditing of vision systems for demographic bias. The Monk Skin Tone scale (Schumann et al., 2023) provides a perceptually uniform basis for skin tone stratification. In generative models, bias may manifest as differential synthesis quality (Cho et al., 2023); DS-VTON (Sun et al., 2025) introduces skin-aware refinement but provides no stratified results.

African Fashion in Computer Vision. AFRIFASHION1600 (Oyewusi et al., 2021) targets classification, not generation; AFRIGAN (Salami et al., 2021) addresses unconditional texture synthesis. Omotayo et al. (2024) find that fewer than 0.06% of CV papers address African visual domains, reflecting the near-absence of African institutions from benchmark construction.

3. Methodology

3.1. Benchmark Construction

AfriVTON-Bench is a curated evaluation dataset designed to expose distribution-shift failures in VTON systems trained

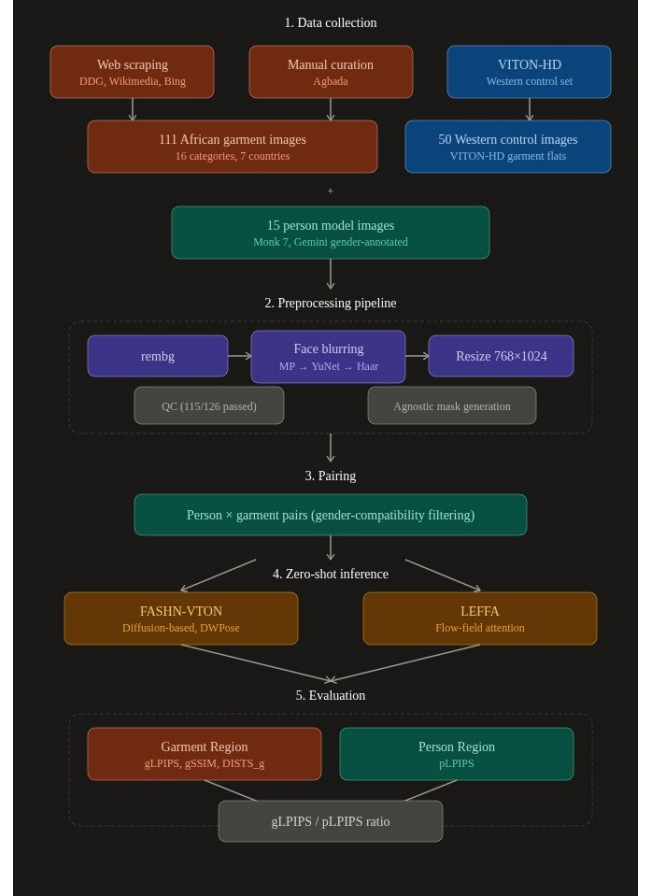


Figure 1. Overview of the AfriVTON-Bench pipeline. Garments are collected through web scraping and manual curation and paired with person model images through gender-compatibility filtering, before zero-shot inference with FASHN-VTON and LEFFA.

on Western fashion corpora. We assembled 111 garment images spanning 16 categories across 7 African countries and 5 geographic sub-regions: Nigeria (Ankara, Agbada, Aso-Oke, Lace), Ghana (Kente, Batakari), Senegal (Boubou, Kaftan), Ethiopia (Habesha Kemis, Netela), Kenya (Kikoi, Shuka), South Africa (Shweshwe), and Morocco (Kaftan, Djellaba). Images were sourced through automated web scraping via DuckDuckGo, Wikimedia Commons, and Bing using the iCrawler library, supplemented by targeted manual curation. A Western control set of 50 garment images was drawn from VITON-HD (Choi et al., 2021). Each garment was annotated with country of origin, geographic region, garment type, visual complexity tier, pattern type, and drape type.

Fifteen person model images were sourced targeting diverse African subjects, annotated with Monk Skin Tone Scale values (Schumann et al., 2023), predominantly Monk 7, and gender labels.

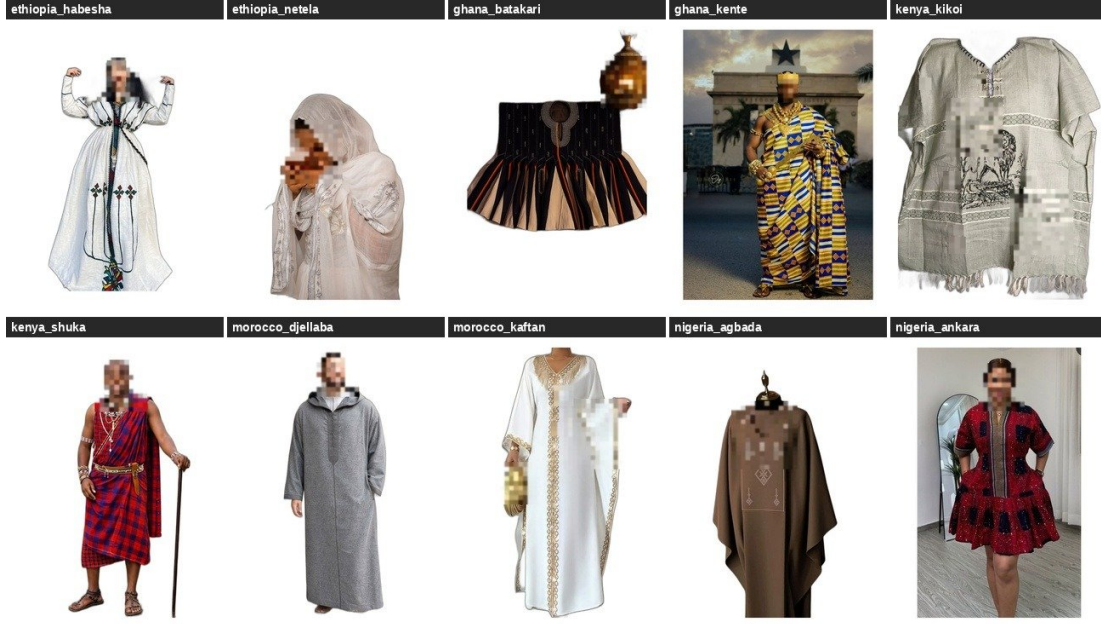


Figure 2. Representative samples from the 16 African garment categories (left to right, top to bottom): Ethiopia Habesha Kemis, Ethiopia Netela, Ghana Batakari, Ghana Kente, Kenya Kikoi, Kenya Shuka, Morocco Djellaba, Morocco Kaftan, Nigeria Agbada, Nigeria Ankara. All images preprocessed to 768×1024 with white backgrounds; faces blurred where present.

3.2. Preprocessing and Inference

All images underwent standardised preprocessing: background removal via `rembg`, padding to 768×1024 , and a cascading multi-detector face blurring pipeline (MediaPipe, YuNet, Haar Cascade) to protect personally identifiable information.

Two VTON architectures were selected to represent distinct design paradigms. **FASHN-VTON** is a diffusion-based model built on Stable Diffusion, employing DWPose for pose estimation (50 denoising timesteps, guidance scale 2.5, seed 42). **LEFFA** (Learning Flow Fields in Attention) learns spatial warping within the attention mechanism of a diffusion model. All experiments ran on an NVIDIA Tesla T4 GPU. FASHN-VTON processed 792 outputs; LEFFA yielded 220 outputs via reduced sampling. The full evaluation set comprises 1,012 try-on images.

3.3. Evaluation Metrics

Four perceptual metrics were computed at image level on 768×1024 images. **Garment LPIPS** (gLPIPS) (Zhang et al., 2018) measures perceptual distance between the try-on output and the reference garment (AlexNet backbone; lower is better). **Garment SSIM** (gSSIM) (Wang et al., 2004) captures luminance, contrast, and structural correspondence (higher is better). **DISTS_g** (Ding et al., 2022) is more sensitive to texture complexity than SSIM. **Person LPIPS** (pLPIPS) quantifies preservation of body, pose, and non-garment appearance. The gLPIPS-to-pLPIPS ratio

Table 1. Aggregate evaluation metrics across all 1,012 person-garment pairs. The g/p Ratio (gLPIPS / pLPIPS) quantifies reconstruction asymmetry; higher values indicate proportionally greater garment texture failure.

Model	gLPIPS↓	gSSIM↑	DISTS _g ↓	pLPIPS↓	g/p
FASHN-VTON	0.650	0.586	0.312	0.179	$3.63 \times$
LEFFA	0.664	0.577	0.312	0.156	$4.27 \times$

serves as a model-agnostic index of garment-specific failure.

4. Results

4.1. Aggregate Performance

Table 1 shows pronounced asymmetry between garment and person reconstruction. FASHN-VTON preserves person identity $3.63 \times$ more faithfully than garment texture; LEFFA’s gap is wider at $4.27 \times$. Despite representing fundamentally different design paradigms, both architectures converge on near-identical garment distortion (gLPIPS 0.650 vs. 0.664; DISTS_g 0.312 vs. 0.312), pointing toward the shared training distribution as the root cause.

4.2. African vs. Western Garments

African garments score 9.4–9.6 percentage points lower in gSSIM than Western controls across both models (Table 2). The gLPIPS values are broadly comparable across origins (≈ 0.650), yet the spatial organisation and geometric regu-

Table 2. Performance stratified by garment origin. The gSSIM gap between African and Western garments is 9.4–9.6 percentage points across both models.

Model	Origin	n	gLPIPS↓	gSSIM↑	pLPIPS↓
FASHN-VTON	African	474	0.650	0.549	0.204
	Western	318	0.651	0.643	0.142
LEFFA	African	142	0.666	0.543	0.167
	Western	78	0.659	0.639	0.134

Table 3. Per-category results ranked by FASHN-VTON gSSIM (ascending). † marks LEFFA cells with fewer than 3 samples.

Category	FASHN-VTON		LEFFA	
	gLPIPS	gSSIM	gLPIPS	gSSIM
Ghana Kente	0.654	0.460	0.674	0.437
Nigeria Ankara	0.711	0.474	0.710	0.477
Ghana Batakari	0.715	0.513	0.677	0.532
Kenya Kikoi	0.717	0.519	0.728	0.495
Kenya Shuka	0.657	0.542	0.729	0.504†
Nigeria Aso-Oke	0.664	0.542	0.629	0.574
Nigeria Lace	0.614	0.561	0.782	0.381†
Morocco Djellaba	0.618	0.575	0.632	0.545†
Nigeria Agbada	0.656	0.589	0.654	0.606
Ethiopia Netela	0.665	0.596	0.672	0.574
S.A. Shweshwe	0.586	0.621	0.601	0.625
Senegal Boubou	0.584	0.623	0.608	0.624
Senegal Kaftan	0.608	0.632	0.638	0.643
Morocco Kaftan	0.566	0.640	0.555	0.655
Ethiopia Habesha	0.572	0.658	0.612	0.646
Western Control	0.651	0.640	0.658	0.638

larity of African garments are disproportionately destroyed during synthesis.

4.3. Per-Category Analysis

Table 3 reveals a failure hierarchy that maps predictably onto visual complexity. Kente and Ankara occupy the bottom two positions in gSSIM for both models, representing a 17–19 percentage point deficit relative to the Western control. The rank ordering is highly consistent across models (Spearman $\rho \approx 0.88$). Notably, Ethiopia Habesha Kemis and Morocco Kaftan cluster at or above the Western control in gSSIM, confirming that degradation is driven by garment visual complexity, not by African origin as such.

4.4. Systematic Failure Modes

Three failure modes are observed consistently across both architectures (see Figure 3):

Pattern dissolution is the most severe failure mode, affecting high-frequency repeating motifs in Ankara wax-print, Kente narrow-strip weave, and Aso-Oke hand-woven patterns. Both models smear or homogenise the motifs during

warping, producing outputs in which the textile’s visual identity—and with it, its cultural significance—is substantially lost.

Drape collapse affects voluminous non-tailored garments such as Agbada and Boubou. Both models collapse flowing multi-layered silhouettes inward into narrower, body-congruent outlines, reflecting an implicit prior over garment shape that is incompatible with silhouettes whose intended appearance depends on free-hanging volume.

Tonal drift is observed in outputs for Monk Scale 7 subjects as subtle skin-tone shifts at garment–body boundaries, where dark skin meets high-contrast textile patterns. The artefact likely reflects inconsistencies in the compositing of the synthesised garment region with the preserved person region during inpainting-based try-on.

5. Cultural Implications

The results converge on a single root cause: the Western centrality of training data underlying current VTON systems. That the degradation is architecture-agnostic—replicated across a DWPose-conditioned diffusion model and an attention-flow warping model—rules out model-level explanations and directly implicates the data distribution.

From a cultural AI perspective, the failure modes we document are not merely technical shortcomings but instances of *cultural erasure through optimisation*. Pattern dissolution destroys the visual identity of textiles that carry deep cultural meaning: Kente cloth encodes proverbs through its Adinkra symbol motifs, Ankara wax-prints function as markers of social occasion and status, and Aso-Oke weaving traditions represent intergenerational craft knowledge. When a VTON model smears these patterns into undifferentiated colour fields, it does not simply reduce an image quality metric—it eliminates the semiotic content that makes these garments culturally meaningful.

Drape collapse similarly imposes a Western body-congruent norm on garment silhouettes whose cultural function depends on volume, layering, and off-body projection. The Agbada’s flowing triple-layer structure signals formality and status in West African social contexts; collapsing it into a fitted outline is a form of cultural normalisation.

These observations support the workshop’s framing of a *positive vision for cultural AI*. Rather than treating cultural diversity solely as a fairness constraint to satisfy, AfriVTON-Bench demonstrates that culturally grounded evaluation can serve as a principled stress test that surfaces generalisation failures of broader relevance. African garments are not a niche edge case but a domain where the collision of high-frequency texture, non-standard silhouette, and darker skin tones exposes fundamental limitations in current generative

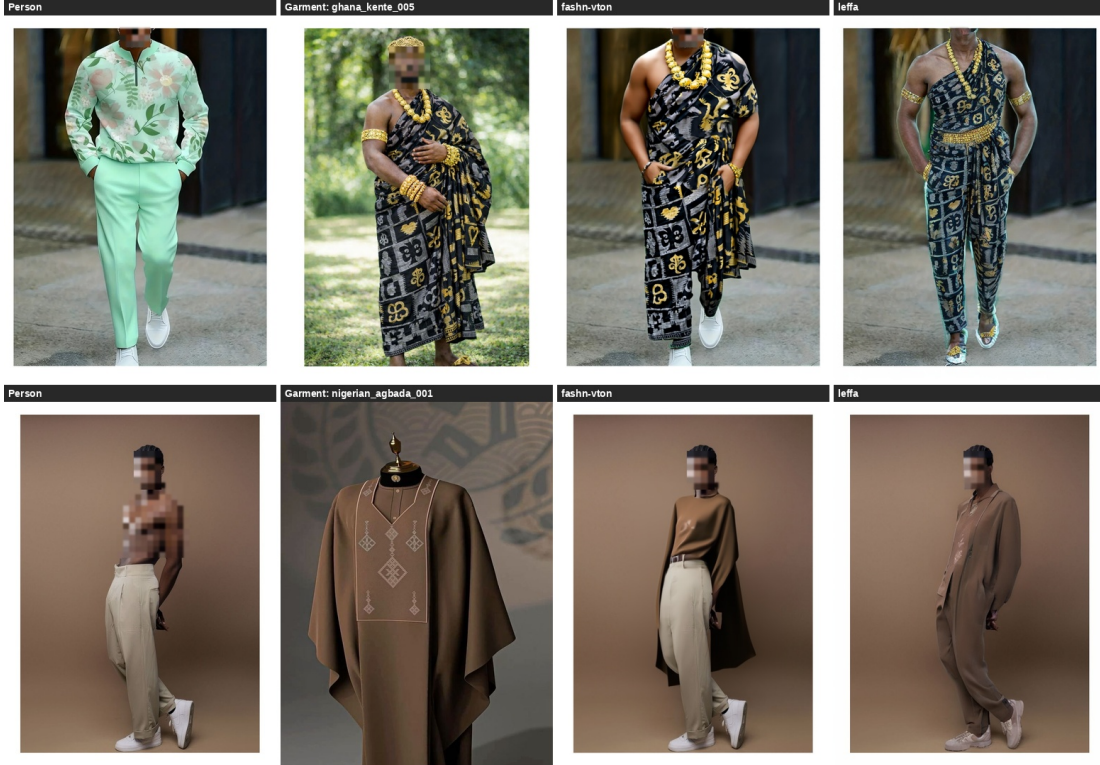


Figure 3. Qualitative comparison illustrating two key failure modes. Each row shows (left to right) the input person, the reference garment, the FASHN-VTON output, and the LEFFA output. Top: Ghana Kente (pattern dissolution—the high-frequency Adinkra motifs are smeared; LEFFA additionally collapses the draped silhouette into fitted trousers). Bottom: Nigeria Agbada (drape collapse—the voluminous triple-layer garment is compressed into a body-congruent outline). Faces blurred for privacy.

architectures. Addressing these limitations would improve system robustness across the full range of global garment diversity.

As global e-commerce platforms expand into African markets, VTON systems that perform poorly on locally worn garments risk providing a substantially degraded shopping experience for African consumers. AfriVTON-Bench demonstrates that benchmark diversity is a prerequisite for trustworthy capability assessment in generative vision, and that systems which appear state-of-the-art on existing benchmarks may fail substantially on garment types worn by the majority of the world’s population.

6. Conclusion

AfriVTON-Bench is the first benchmark designed to evaluate VTON robustness on African garment types and skin tone diversity. Evaluating FASHN-VTON and LEFFA across 1,012 person–garment pairs reveals systematic, architecture-agnostic performance degradation: a 9–10 point gSSIM deficit on African garments relative to Western controls, with Kente and Ankara exhibiting deficits of 17–19 points. Three failure modes—pattern dissolution, drape collapse, and tonal drift—are all attributable to the Western

centricity of existing VTON training corpora.

For the Culture × AI community, this work offers a concrete case study of how culturally situated benchmarks can move beyond harm documentation toward actionable system improvement. The dataset, evaluation code, and model outputs will be released publicly to support reproducible research on culturally inclusive garment synthesis.

Ethical Considerations. All person images were processed through a multi-detector face blurring pipeline; no face is visible in any released image. Our person image set is small and skewed toward Monk 7, limiting tonal drift conclusions; future versions will include a broader demographic range. AI-assisted writing tools were used during manuscript preparation; all content is the original work of the authors.

Limitations. The primary limitation is the size of the person image set (15 images, predominantly Monk Scale 7). Evaluation is restricted to two architectures and lacks user study data for perceptual ground truth from human evaluators.

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